Citi Bike Rebalancing With RL and Heuristic Baselines

March 2026

Problem

- Empty origin stations create unmet trips.
- Full destination stations create overflow and operational waste.
- Rebalancing is sequential: moving bikes now can help later demand, but every move has a cost.
- The project goal is to test whether RL can beat a strong forecast-based heuristic under a realistic temporal holdout.

Research Question

- Can RL outperform a demand-aware heuristic on a strict future-month test set?
- Why this matters: beating a no-op baseline is easy to overstate, but beating a forecast-aware operational baseline is meaningful.
- Final evaluation: train on January 2025 through January 2026 and test on February 2026.

Environment Setup

- Five-station Jersey City subproblem: `JC115`, `HB101`, `HB106`, `JC009`, `JC109`
- State: inventory, calendar, holiday/weather, and demand-profile features
- Actions: `no_op` plus fixed-size directed bike transfers
- Reward: served-trip reward minus unmet-demand, move, and overflow penalties

Data

- Citi Bike Jersey City monthly trip files from January 2025 through February 2026
- NOAA daily weather from station `USW00014734`
- 14 processed monthly flow files

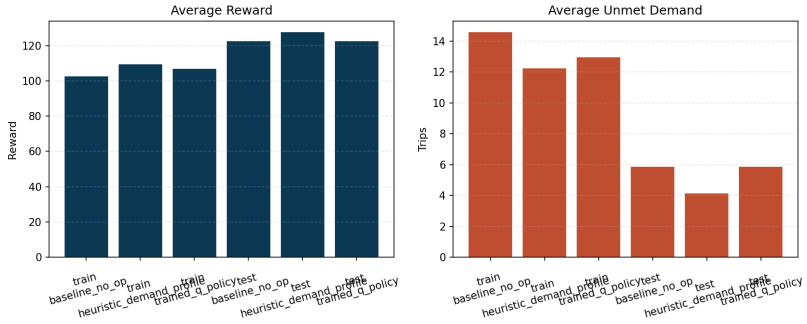
- 423 daily demand episodes total
- 396 train episodes and 27 February 2026 test episodes in the primary holdout

Methods

- `baseline_no_op`
- `heuristic_demand_profile`
- Tabular Q-learning progression: inventory only, then calendar, forecast, holiday, and weather context
- Dueling Double DQN with dense state features
- DQN move-margin regularization: only move when `Q(move) >= Q(no_op) + margin`

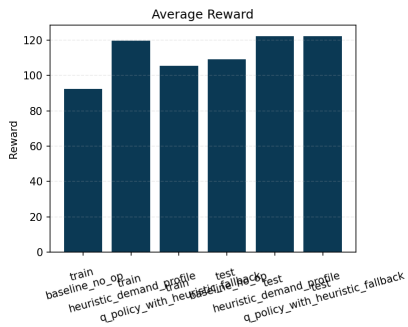
Development Result

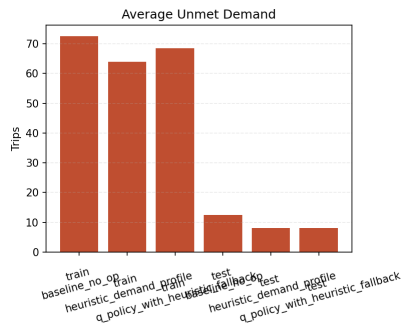
- On the early within-February split, the heuristic immediately became the real benchmark.
- Calendar features helped tabular Q-learning close the gap to no-op.
- But the demand-profile heuristic still won clearly.



Main Seasonal Holdout Result

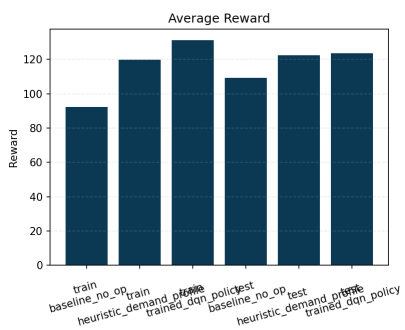
- Strict holdout: train through January 2026, test on February 2026
- No-op baseline: `109.33`
- Heuristic baseline: `122.45`
- Q table with heuristic fallback: `122.21`
- The Q-table is almost entirely fallback on this holdout: `23.96` fallback actions per day

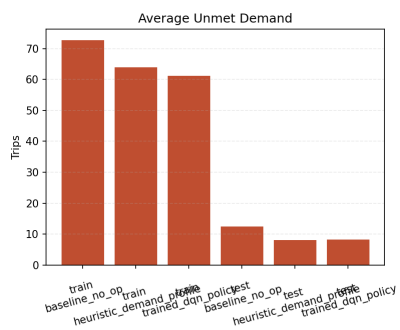




DQN Result and Caveat

- Unregularized DQN over-moved bikes and underperformed the heuristic.
- Move-margin regularization improved the best DQN run: `123.63` vs heuristic `122.45`
- But that win was not stable across seeds.





Robustness Check

Seed	DQN avg reward	Heuristic avg reward
7	123.63	122.45
11	113.61	122.45
19	116.95	122.45

- Conclusion: the regularized DQN is promising, but not robust enough to claim a stable improvement.

Takeaways

- The repo now contains a complete, reproducible RL research pipeline with real demand and weather data.
- The demand-profile heuristic is the strongest robust policy on the February 2026 holdout.
- The move-margin DQN best run is encouraging, but stability is still the main blocker.
- Next modeling step: residual learning around the heuristic, plus broader seed and holdout evaluation.